

#HW 9

● Graded

Student

Nick Labree

Total Points

97 / 100 pts

Question 1

(no title)

15 / 15 pts

✓ - 0 pts Correct

Question 2

(no title)

15 / 15 pts

✓ - 0 pts Correct

- 1 pt Mass flow rate should be around 4.9 kg/s.

- 1 pt Area should be around 0.07 sq. m

Question 3

(no title)

14 / 15 pts

- 0 pts Correct

✓ - 1 pt Power should be around 3500 kW.

Question 4

(no title)

14 / 15 pts

- 0 pts Correct

- 3 pts Incomplete

✓ - 1 pt Heat transfer rate will be around -15 kW.

Question 5

(no title)

20 / 20 pts

✓ - 0 pts Correct

- 1 pt POWER=0.52kW

- 4 pts Incomplete

Question 6

(no title)

19 / 20 pts

– 0 pts Correct

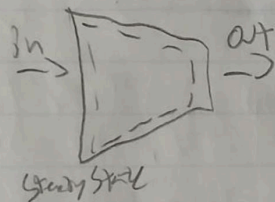
✓ – 1 pt $x=0.191$

– 6 pts incomplete

No questions assigned to the following page.

CHE HW 9 Nicklas

1. Refrigerant 134A enters a well-insulated nozzle at 200 lbf/in², 220°F, with a velocity of 120 ft/s and exits at 20 lbf/in² with a velocity of 1500 ft/s. For steady state operation and neglecting potential energy difference effects



Open System

$$\dot{m}_1 = \dot{m}_2 = \dot{m}$$

$$\dot{m} \left(h_1 + \frac{V_1^2}{2g_c} \right) = \dot{m} \left(h_2 + \frac{V_2^2}{2g_c} \right)$$

$$P_1 = 200 \text{ psia} = 1.4 \text{ MPa} \quad T_1 = 220^\circ\text{F} = 104.4^\circ\text{C}$$

$$\frac{h_1 - h_2}{T_1 - T_2} = \frac{h_1 - h_2}{T_1 - T_2} = \frac{41.2 - 330.3}{110 - 100} = \frac{X - 330.3}{104.4 - 100} \quad h_1 = 335 \text{ kJ/kg} \quad T_1 = 220^\circ\text{F}$$

$$\Delta KE = \frac{V_2^2 - V_1^2}{2g_c} = \frac{1500^2 - 120^2}{2(32.174)}$$

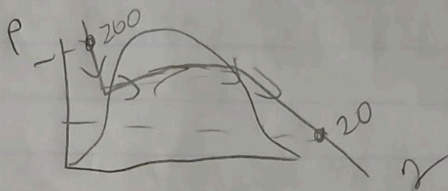
$$\Delta KE = \frac{3.174 \times 10^6}{778} = 4074 \text{ Btu/lbm}$$

No questions assigned to the following page.

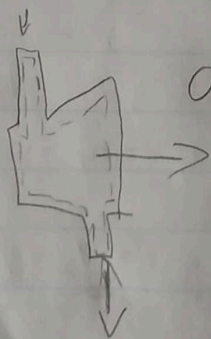
$$h_2 = h_1 - \Delta KE = 144 - 44.7 \approx 99.3 \text{ BTU/lbm}$$

$$20 \text{ psia } R = 134 \text{ ft}^2/\text{lbm}^2 \cdot \text{R}^2 \approx 10 \quad T_2 = T_{\text{sat}}(20 \text{ psia})$$

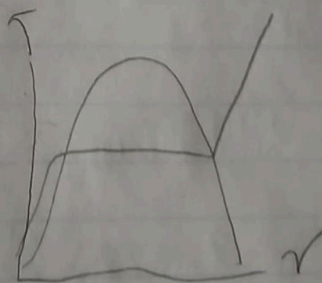
$$-2.5^\circ \text{F}$$



2. Air expands through a turbine from 8 bar
960 K to 1 bar/450 K. The inlet velocity is
small compared to the exit velocity of 90 m/s.
The turbine operates at steady state and develops
a power output of 2500 kW. Heat transfer
between the turbine and its surroundings and potential
energy effects are negligible. model air as an
ideal gas?



Open system



No questions assigned to the following page.

2.

$$\dot{m}_1 = \dot{m}_2 = \dot{m}$$

$$\dot{Q} - \dot{W}_s + \dot{m} \left(h_1 + \frac{V_1^2}{2} + g z_1 \right) = \dot{m} \left(h_2 + \frac{V_2^2}{2} + g z_2 \right)$$

$$\dot{Q} = 0 \quad g z_1 \approx g z_2 \approx 0 \quad V_1 = 0$$

$$\dot{W}_s = \dot{m} \left[(h_1 - h_2) - \frac{V_2^2}{2} \right]$$

$$h_1 - h_2 = c_p (T_1 - T_2) = 1.004 (960 - 450)$$

$$= 1.004 \times 510 = 512 \text{ kJ/kg}$$

$$\frac{V_2^2}{2} = \frac{(10 \text{ m/s})^2}{2} = \frac{100}{2} = 50 \text{ J/kg} = 0.05 \text{ kJ/kg}$$

$$\dot{W}_s = (h_1 - h_2) - \frac{V_2^2}{2} = 512 - 0.05 = 508 \text{ kJ/kg}$$

$$\dot{m} = \frac{\dot{W}_s}{w_s} = \frac{2500 \text{ kJ/s}}{508 \text{ kJ/kg}} = 4.92 \text{ kg/s}$$

$$\dot{m} = 4.92 \text{ kg/s}$$

$$\dot{m} = \rho_2 V_2 A_2 \Rightarrow A_2 = \frac{\dot{m}}{\rho_2 V_2}$$

$$P_2 = 1 \text{ bar} = 100 \text{ kPa}$$

$$T_2 = 450 \text{ K}$$

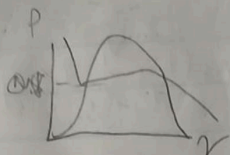
$$R_{\text{air}} = 0.287 \text{ kJ/kg} \cdot \text{K}$$

$$\rho_2 = \frac{P_2}{R T_2} = \frac{100000 \text{ Pa}}{0.287 \times 450} = \frac{100000}{129150} = 0.775 \text{ kg/m}^3$$

$$A_2 = \frac{4.92}{0.775 \times 10} = \frac{4.92}{7.75} \approx 0.070 \text{ m}^2$$

No questions assigned to the following page.

3



Steady state
Water vapor behaves
as ideal steam
adiabatic turbine
 $\dot{Q} \approx 0$
 $\Delta E \approx 0$

$$\dot{m}_1 = \dot{m}_2 = \dot{m}$$

$$\dot{Q} - \dot{W}_s + \dot{m} \left(h_1 + \frac{V_1^2}{2} + gz_1 \right) = \dot{m} \left(h_2 + \frac{V_2^2}{2} + gz_2 \right)$$

$$\dot{W}_s = \dot{m} \left[(h_1 - h_2) + \frac{V_1^2 - V_2^2}{2} \right]$$

$$V_2 = 9.48 \text{ m}^3/\text{s}$$

$$p_2 = 0.8 \text{ bar} = 80 \text{ kPa}$$

$$T_{\text{sat}} = 94^\circ\text{C}$$

$$V_1 = 2.58 \text{ m}^3/\text{kg} \quad \dot{m} = \frac{V_2}{V_2} = \frac{9.48}{2.58} = 3.67 \text{ kg/s}$$

$$0.8 \text{ bar} \rightarrow h_2 = h_g(0.8 \text{ bar}) \approx 2680 \text{ kJ/kg}$$

40 bar 500C

$$h_1 = 3445 \text{ kJ/kg}$$

$$h_1 - h_2 = 3445 - 2680 = 765 \text{ kJ/kg}$$

$$\frac{V_1^2 - V_2^2}{2} = \frac{20^2 - 150^2}{2} = \frac{4000 - 22500}{2} = \frac{-17500}{2}$$

$$= -8750 \text{ J/kg} = -8.75 \text{ kJ/kg}$$

No questions assigned to the following page.

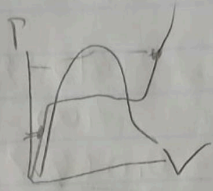
3

$$W_s = (h_1 - h_2) + \frac{V_1^2 - V_2^2}{2} = 765 + 8.75 \approx 773.75 \text{ kJ/kg}$$

$$\dot{W}_s = \dot{m} W_s = 3.67 \text{ kg/s} \cdot 773.75 \text{ kJ/kg} = 2839.75 \text{ kW}$$

$$\dot{W}_s = 2.84 \text{ MW}$$

4.



Steady State

Refrigerant R-134a
Use superheated
Vapor properties

kinetic and potential energy effects negligible

No chemical reactions

$$\dot{W}_s = 60 \text{ kJ/kg}$$

$$\dot{m}_1 = \dot{m}_2 = \dot{m} \quad \dot{Q} - \dot{W} = \dot{m}(h_2 - h_1)$$

$$\dot{W} = -\dot{m} W_h = -\dot{m} (60 \text{ kJ/kg})$$

$$\dot{Q} - (-\dot{m} 60) = \dot{m}(h_2 - h_1)$$

$$\dot{Q} = \dot{m} [(h_2 - h_1) - 60]$$

No questions assigned to the following page.

$$V_1 = 4 \text{ m}^3/\text{min} = \frac{4}{60} \text{ m}^3/\text{s} = 0.0667 \text{ m}^3/\text{s}$$

$$R_{134a} \approx 0.0815 \text{ kJ/kg} \cdot \text{K} = 0.0815 \text{ kPa} \cdot \text{m}^3/\text{kg} \cdot \text{K}$$

$$T_1 = 20^\circ\text{C} = 293 \text{ K} \quad p_1 = 4 \text{ bar} = 400 \text{ kPa}$$

$$v_1 = \frac{R T_1}{p_1} = \frac{0.0815 \times 293}{400} = 0.0597 \text{ m}^3/\text{kg}$$

$$\dot{m} = \frac{\dot{V}_1}{v_1} = \frac{0.0667}{0.0597} \approx 1.12 \text{ kg/s}$$

$$T_1 = 20^\circ\text{C} \quad T_2 = 80^\circ\text{C} \quad c_p = 0.9 \text{ kJ/kg} \cdot \text{K}$$

$$h_2 - h_1 = c_p (T_2 - T_1) = 0.9 \times (80 - 20) = 0.9 \times 60 = 54 \text{ kJ/kg}$$

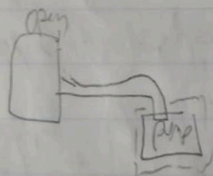
$$q = (h_2 - h_1) - 60 = 54 - 60 = -6 \text{ kJ/kg}$$

$$\dot{Q} = \dot{m} q = (1.12 \text{ kg/s})(-6 \text{ kJ/kg}) = -6.7 \text{ kW}$$

$$\dot{Q} = -6.7 \text{ kW}$$

Steady state inlet/exist

5)



$$p \approx 1000 \text{ kg/m}^3 \quad \text{at } 15^\circ\text{C}$$

Temp constant

1 share internal energy change

Heat transfer negligible

$$\dot{Q} \approx 0$$

ke negligible

pump shaft work is the only work interaction

No questions assigned to the following page.

4

ND

$$\dot{m}_1 = \dot{m}_2 = \dot{m}$$

$$\dot{Q} - \dot{W}_s + \dot{m} \left(h_1 + \frac{V_1^2}{2} + g z_1 \right) = \dot{m} \left(h_2 + \frac{V_2^2}{2} + g z_2 \right)$$

$$\dot{Q} = 0 \quad V_1 = V_2 \quad h_2 - h_1 = \frac{p_2 - p_1}{\rho}$$

$$- \dot{W}_s + \dot{m} \left(\frac{p_1}{\rho} + g z_1 \right) = \dot{m} \left(\frac{p_2}{\rho} + g z_2 \right)$$

$$\dot{W}_s = \dot{m} \left[\frac{p_2 - p_1}{\rho} + g (z_2 - z_1) \right]$$

$$\frac{\Delta p}{\rho} = \frac{2.0 \times 10^5}{1000} = 200 \text{ J/kg}$$

$$g \Delta z = 9.81 \times 15 = 147.15 \text{ J/kg}$$

$$w_{\text{pump}} = 200 + 147.15 = 347.15 \text{ J/kg}$$

$$\dot{W}_m = \dot{m} w_{\text{pump}} = 1.5 \times 347.15 = 520.7 \text{ W} = 0.52 \text{ kW}$$

$$\dot{W}_m = 0.52 \text{ kW}$$

No questions assigned to the following page.

6)



Steady state, single/shale exit
Throttling (Flow through a restriction).

$$\dot{Q}=0, \dot{W}=0$$

ΔKE & ΔPE Negligible

$$h_1 = h_2$$

10 bar 25°C $24^\circ\text{C} \rightarrow$ Slightly
Subcooled/compressed
liquid

$$\dot{m}_1 = \dot{m}_2 = \dot{m}$$

$$\dot{Q} - \dot{W} + \dot{m} \left(h_1 + \frac{V_1^2}{2} + gz_1 \right) = \dot{m} \left(h_2 + \frac{V_2^2}{2} + gz_2 \right)$$

$$\dot{m} \left(\dot{Q}=0, \dot{W}=0, \Delta KE=0, \Delta PE=0 \right)$$

$$\dot{m} h_1 = \dot{m} h_2 \quad h_1 = h_2$$

Inlet enthalpy 10 bar Sat. Temp = 24.8°C
460 kJ/kg 24 C

$$h_1 = h_f(10 \text{ bar}) = 460 \text{ kJ/kg}$$

exit pressure $p_2 = 1 \text{ bar}$ at 1 bar ammonia saturated
at about -33.66°C

$$h_f(1 \text{ bar}) = 190.8 \text{ kJ/kg} \quad h_g(1 \text{ bar}) = 682.3 \text{ kJ/kg}$$

from steam Table, CO

at 1 bar, the state with h_2 between h_f and h_g
is a mixture

No questions assigned to the following page.

availability $x_2 = \frac{h_2 - h_6}{h_5 - h_f} = \frac{460 - 190.8}{682.3 - 190.8} = \frac{269.2}{491.5}$

≈ 0.55

T

